Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the OptiCrop application can be enhanced to support more users, larger datasets, and additional intelligent features in the future. It also presents the planned improvements to increase the application's performance, usability, scalability, and deployment readiness.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Works only on a local Flask server	Cannot be accessed by users over the Internet	High
2	Uses a single crop recommendation dataset	Limited recommendations for different regions	Medium
3	Supports only crop recommendation	Does not provide fertilizer or disease suggestions	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on a cloud platform to support multiple users simultaneously
2	Data Storage	Uses CSV dataset	Store agricultural data in a relational database such as MySQL
3	Performance	Single prediction model	Optimize the model and implement caching for faster predictions
4	Security	Basic local application	Add user authentication, HTTPS, and secure input validation

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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy the application on the cloud	1–2 Months	Public access from anywhere
Phase 2	Add fertilizer recommendation	2–3 Months	Better decision support for farmers
Phase 3	Add plant disease detection using image classification	3–5 Months	Improve crop health management
Phase 4	Develop an Android mobile application with multilingual support	5–6 Months	Increase accessibility and usability for farmers